MSA Visual Posters - Quality Classroom and Shop Floor

A3 reference posters for MSA concepts | AIAG MSA 4th Edition

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

POSTER 1: GRR Decision Tree

- %GRR < 10%: ACCEPT - Gauge approved for process control and inspection
- %GRR 10-30%: CONDITIONAL - Evaluate gauge cost vs risk. Approved only with Quality Manager sign-off.
- %GRR > 30%: REJECT - Gauge must be improved or replaced. Do NOT use for CC characteristics.
- NDC < 5: FAIL - Even if %GRR is acceptable, NDC < 5 means gauge cannot distinguish parts.
- ALWAYS check BOTH: % Study Variation AND % Tolerance before approving any gauge.

POSTER 2: MSA Study Types Quick Reference

- GRR (Variable): For all variable gauges | Method: 10 parts x 3 operators x 2 trials | Metric: %GRR, NDC
- Bias: For all reference/precision gauges | Method: 25 measurements of master | Metric: %Bias, t-test
- Linearity: For gauges used across a range | Method: 5 ref parts x 12 readings | Metric: R-squared, %Linearity
- Stability: Ongoing for critical gauges | Method: Control chart, 3 reads/period | Metric: UCL/LCL in control
- Attribute GRR: For go/no-go and visual | Method: 50 parts x 3 appraisers x 2 trials | Metric: Kappa coefficient

POSTER 3: Calibration vs GRR - Key Difference

- CALIBRATION: Compares gauge reading to a certified reference standard. Adjusts zero/span. Done by metrology.
- GRR: Measures the random variation in the gauge when used in production conditions. Done with real operators and real parts.
- A CALIBRATED gauge can STILL FAIL GRR - if worn, used incorrectly, or operator-sensitive.
- RULE: Always do GRR AFTER calibration. Calibration first, then GRR. Not the other way around.
- Auditor will check BOTH - calibration certificate AND GRR study report. Having only one is a finding.

POSTER 4: MSA Before PPAP - Checklist

- [ ] GRR study completed for all measurement systems in Control Plan
- [ ] GRR result < 30% (target < 10% for CC characteristics)
- [ ] NDC >= 5 for all variable gauges
- [ ] Bias study completed for all reference gauges
- [ ] Linearity study completed for gauges used across range
- [ ] Stability monitoring plan in place (especially for CC gauges)
- [ ] Attribute GRR completed for all go/no-go and visual inspection points
- [ ] All MSA results documented in PPAP Element 8 package
- [ ] Customer-specific MSA requirements reviewed and met